

Mobile Network Quality-of-Service Analysis and Prediction

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Problem Statement

The increasing deployment of 4G and 5G mobile networks has resulted in a rapid growth of data traffic and connected devices. Maintaining consistent Quality of Service (QoS) has become a major challenge due to varying network conditions, congestion, and user mobility.

Telecommunication providers require data-driven techniques to analyze network performance and predict future QoS metrics for efficient network planning and optimization.

Objectives

- Analyze mobile network QoS data.
- Identify factors affecting network performance.
- Visualize QoS trends using statistical methods.
- Predict future QoS metrics using machine learning.
- Develop an interactive dashboard for network monitoring.

Proposed Methodology

Objectives

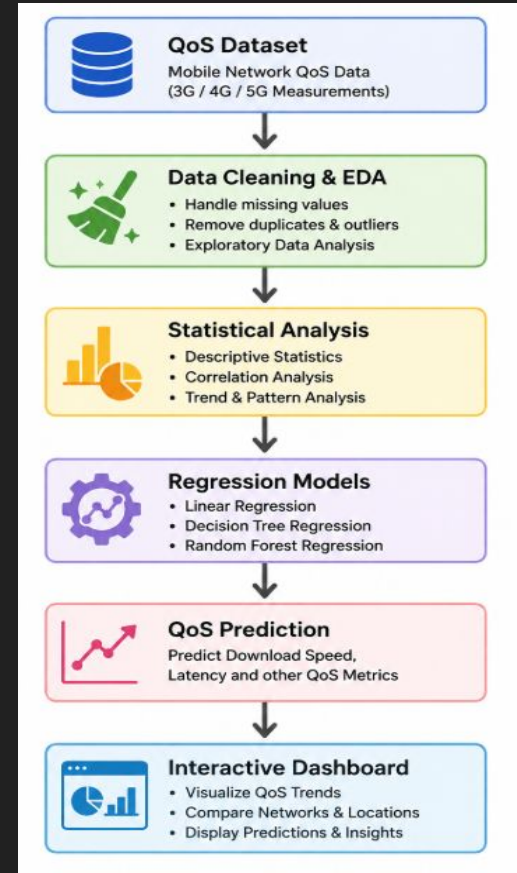
- Analyze historical mobile network QoS data.
- Identify relationships among key QoS metrics.
- Develop predictive models for network performance.
- Build an interactive analytics dashboard.

Key QoS Metrics

- Download Speed
- Upload Speed
- Latency
- Packet Loss
- Jitter
- Signal Strength
- Network Technology (3G/4G/5G)

Expected Outcome

- Better understanding of network behavior.
- Accurate prediction of QoS metrics.
- Interactive visualization for performance monitoring and decision support.



Data Set

Primary Dataset

5G Network Data & Ookla Open Data for Internet Speed

Description

- Combines 5G network QoS metrics with real-world internet speed measurements across multiple locations.
- Includes key parameters such as download speed, upload speed, latency, signal strength, jitter, carrier, and regional performance data.

Why These Datasets?

- Provide comprehensive QoS and network performance indicators.
- Support EDA, statistical analysis, machine learning-based QoS prediction, and interactive dashboard development.
- Closely align with the project's objective of analyzing and predicting mobile network performance.

Thank you